Genomics

In the wake of transformative advances in sequencing technologies, ELIXIR CZ is spearheading efforts to unravel the genomic mysteries of life on Earth. Our commitment aligns with global initiatives like the Earth BioGenome Project, focusing on decoding the DNA of all living organisms. As we delve into this ambitious era, ELIXIR CZ strategically concentrates on key research areas with a rich legacy of excellence in sequencing in the Czech Republic.

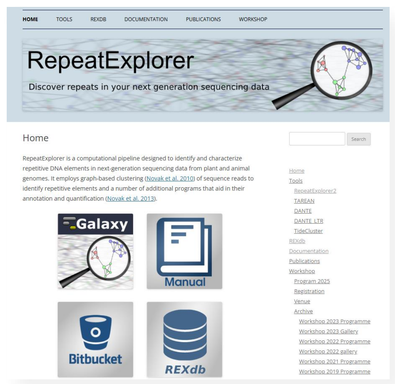
What We Do:

- **Annotation of Repetitive DNA:** Developing tools such as RepeatExplorer and TAREAN to identify and document repetitive DNA elements.
- **Phylogenomics and Barcoding:** Harnessing large-scale genomic data to study evolutionary relationships and support conservation efforts, enhanced by tools like the AmtDB database.
- **Targeted Genomic Studies:** Specializing in the genomics of parasitic protists, neglected crops, and complex microbial communities, backed by resources like the GlobalFungi database.

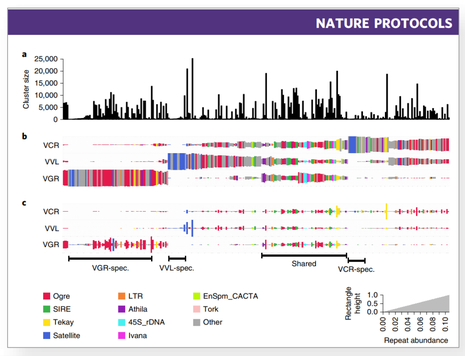
Our Goals:

- **Innovate and Integrate:** Promote the development of cutting-edge computational tools and databases, ensuring their integration into international frameworks – such as **Galaxy platform**.
- **Enhance Support for Genomic Projects:** Provide the necessary hardware and expertise to manage and share the burgeoning volume of genomic data.
- **Democratize Access:** Offer user-friendly tools and training to empower researchers across various disciplines, making advanced genomics accessible to all.

Repeat Explorer

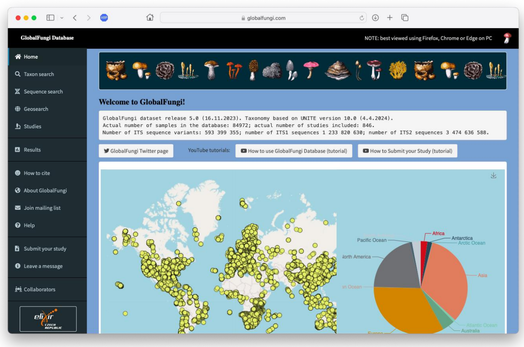


- Computational pipeline to identify and characterize repetitive DNA elements in NGS data from plant and animal genomes
- + REXdb: a reference database of transposable element protein domains

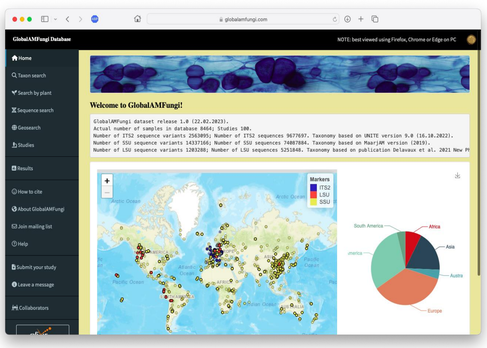


repeatexplorer.org/
repeatexplorer-elixir.cerit-sc.cz

Global Fungi



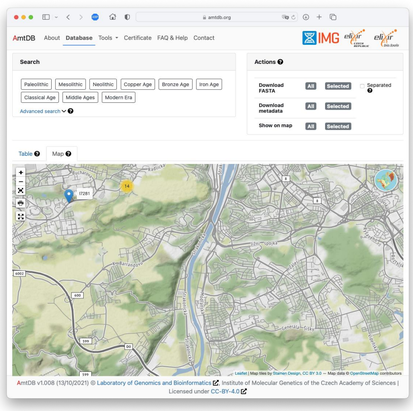
- a comprehensive, open-access database dedicated to documenting the global diversity of fungal communities.
- + GlobalAMFungi - open-access database dedicated to cataloging the global diversity of arbuscular mycorrhizal nsive, open-access fungi (AMF).
- Highly-cited paper



globalfungi.com
globalamfungi.com

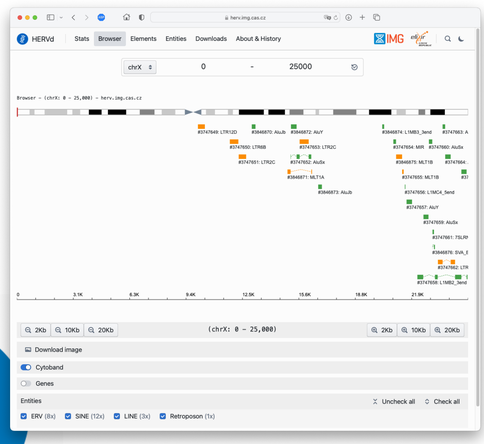
AmtDB

- database of ancient human mitochondrial genomes.
- The first database of ancient human mitochondrial genomes.
- A key platform for ancient population genetic studies
- Contains 2,548 hand-curated ancient samples

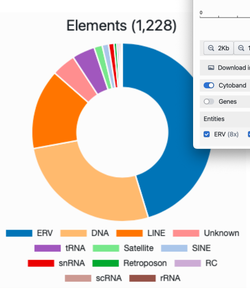


amtdb.org

HERVd



- database of human endogenous retroviruses
- Version 4.2 with almost 4 million distinct entries
- 1228 ERV elements
- Connected to main genome browsers (e.g. ensEMBL)



herv.img.cas.cz

GOLEM

- Gene regulatOry eLEMents
  web server of gene regulatory elements in plants
- Plant genomes across the plant Tree of Life project

golem.ncbr.muni.cz

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